



Full GBA + X/Y Buttons + PWM Audio (User-Confirmed Free Pins)

Free pins: 3,5,7,8,10,12,13,15,16 + 29-40 (15 pins total). 11 buttons (GBA + X/Y) + PWM speaker fit perfectly!

PWM Speaker Module

Module Pin	Physical Pin	BCM GPIO
VCC	Pin 1 (3.3V)	-
GND	Pin 30	-
IN	12	GPIO 18

Complete GBA + X/Y Buttons (11 Buttons)

GBA/XY Button	Physical Pin	BCM GPIO	3.3V	GND (10kΩ)
UP	3	GPIO 2	Pin 1	Pin 30
DOWN	5	GPIO 3	Pin 1	Pin 30
LEFT	7	GPIO 4	Pin 1	Pin 30
RIGHT	8	GPIO 14	Pin 1	Pin 30
A	10	GPIO 15	Pin 1	Pin 30
B	13	GPIO 27	Pin 1	Pin 30
X	15	GPIO 22	Pin 1	Pin 30
Y	16	GPIO 23	Pin 1	Pin 30
SELECT	29	GPIO 5	Pin 1	Pin 30
START	31	GPIO 6	Pin 1	Pin 30
L	35	GPIO 19	Pin 1	Pin 30
R	40	GPIO 21	Pin 1	Pin 30

Spare pins: 32(GPIO12), 33(GPIO13), 36(GPIO16), 37(GPIO26), 38(GPIO20)

Visual Layout (All Free Pins)

Pin 3:	GPI02=UP	✓	FREE
Pin 5:	GPI03=DOWN	✓	FREE
Pin 7:	GPI04=LEFT	✓	FREE
Pin 8:	GPI014=RIGHT	✓	FREE
Pin 10:	GPI015=A	✓	FREE
Pin 12:	GPI018=AUDIO	✓	FREE (Speaker)
Pin 13:	GPI027=B	✓	FREE
Pin 15:	GPI022=X	✓	FREE
Pin 16:	GPI023=Y	✓	FREE
Pin 29:	GPI05=SELECT	✓	FREE
Pin 31:	GPI06=START	✓	FREE
Pin 35:	GPI019=L	✓	FREE
Pin 40:	GPI021=R	✓	FREE

Configuration Files

/boot/config.txt **(PWM Audio)**

```
dtparam=audio=on
gpio=18=a2 # PWM Audio on Pin 12 (GPI018)
```

RetroArch Mapping (/opt/retropie/configs/all/retroarch.cfg)

```
input_player1_up = "2"      # UP (Pin 3)
input_player1_down = "3"    # DOWN (Pin 5)
input_player1_left = "4"    # LEFT (Pin 7)
input_player1_right = "14"  # RIGHT (Pin 8)
input_player1_a = "15"      # A (Pin 10)
input_player1_b = "27"      # B (Pin 13)
input_player1_x = "22"      # X (Pin 15)
input_player1_y = "23"      # Y (Pin 16)
input_player1_select = "5"   # SELECT (Pin 29)
input_player1_start = "6"    # START (Pin 31)
input_player1_l2 = "19"      # L (Pin 35)
input_player1_r2 = "21"      # R (Pin 40)
```

✓ **Full GBA + X/Y buttons (11 total) + PWM audio using ONLY confirmed free pins! Perfect SNES/GBA compatibility.** [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#)

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1. <https://www.hackster.io/news/play-any-retro-game-you-want-with-this-game-boy-advance-and-raspberry-pi-project-eaab30c130ce>
2. [https://www.waveshare.com/wiki/3.5inch_RPi_LCD_\(A\)](https://www.waveshare.com/wiki/3.5inch_RPi_LCD_(A))
3. [https://www.waveshare.com/wiki/3.5inch_RPi_LCD_\(F\)](https://www.waveshare.com/wiki/3.5inch_RPi_LCD_(F))
4. <https://www.waveshare.com/3.5inch-rpi-lcd-a.htm>

5. <https://github.com/Gameboypi/SPW>
6. <https://www.instructables.com/Raspberry-Pi-1-Gameboy/>
7. https://www.reddit.com/r/raspberry_pi/comments/4631y1/gameboy_advance_build_with_a_pi_zero_needed_help/
8. https://www.youtube.com/watch?v=FoJr-W_SFdu
9. https://www.reddit.com/r/raspberry_pi/comments/aafmep/complete_gameboy_advance_sp_with_a_raspberry_pi/
10. <https://www.facebook.com/groups/TheRetroFuture/posts/8426322350802739/>
11. <https://www.slideshare.net/slideshow/raspberry-pi-gpio-tutorial-make-your-own-game-console/168517769>
12. <https://www.zegamamegear.uk>
13. <https://retropie.org.uk/forum/topic/4844/pi-in-a-gameboy-advance-build-wip/62>